VEGA HACKATHON 2026

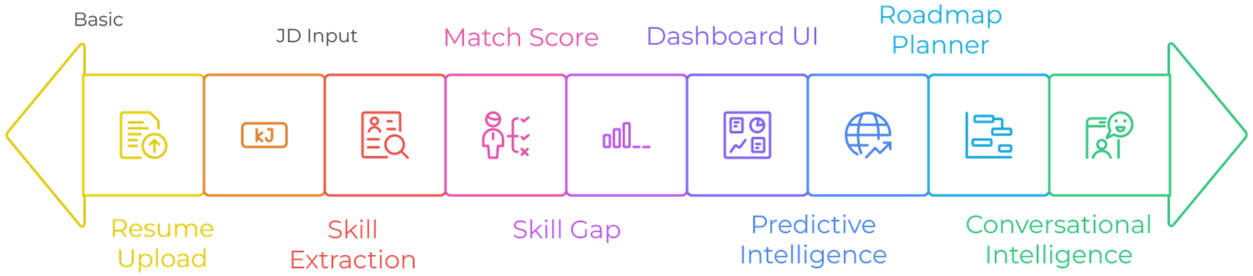
- **Problem Statement ID – AIML-02**
- **Problem Statement Title-** Design and implement an NLP-based intelligent resume analysis system that matches candidate resumes against a given job description and generates Skill match score, Identified core skills, Missing skill gaps, Resume improvement suggestions, Visual comparison dashboard
- **Theme -** EdTech
- **PS Category -** Software
- **Organization :** National Skill Development Corporation (NSDC)
- **Team Name (Registered on portal) -** Singularity

Singularity

Niti.Ai - Your Personalized AI Career Mentor

Proposed solution

AI-powered career mentor that **evaluates job-ready resumes, identifies skill gaps with personalized learning paths**, and delivers **realistic interview simulations with feedback making career planning** structured, efficient, and personalized



Innovation and Uniqueness

- **Predictive Career Intelligence Engine** with market trend benchmarking
- **AI-powered Roadmap Planner** with milestone structuring and dependency.
- **Conversational Resume Intelligence** with semantic rewriting
- **Voice-enabled Mock Interview Simulator** with adaptive questioning and structured feedback

Rohan, 23, uploads his resume and applies for multiple roles. His resume format varies, skills lack depth clarity, and he applies to jobs without meeting minimum criteria. Recruiters manually filter thousands of such applications daily.

Before: Inconsistent resumes, keyword mismatch, manual screening, wasted time.

How it addresses the problem

With Niti.Ai, Rohan's resume is evaluated against the JD - match score is generated, missing skills are identified, and personalized improvement suggestions are provided. Recruiters receive ranked, structured candidate insights.

After: Smart resume evaluation, transparent scoring, skill-gap guidance, faster and better hiring decisions.

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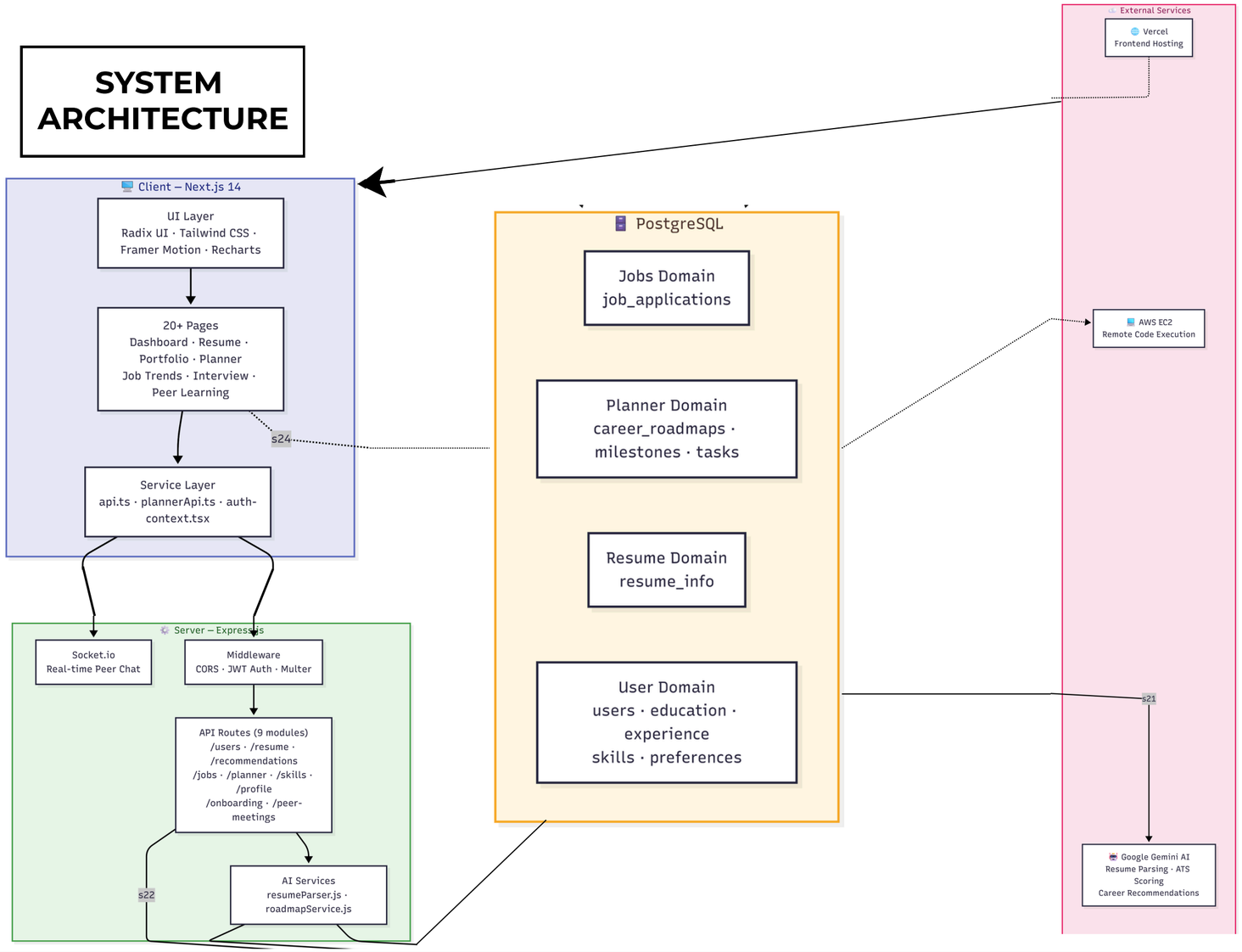
TECHNICAL APPROACH

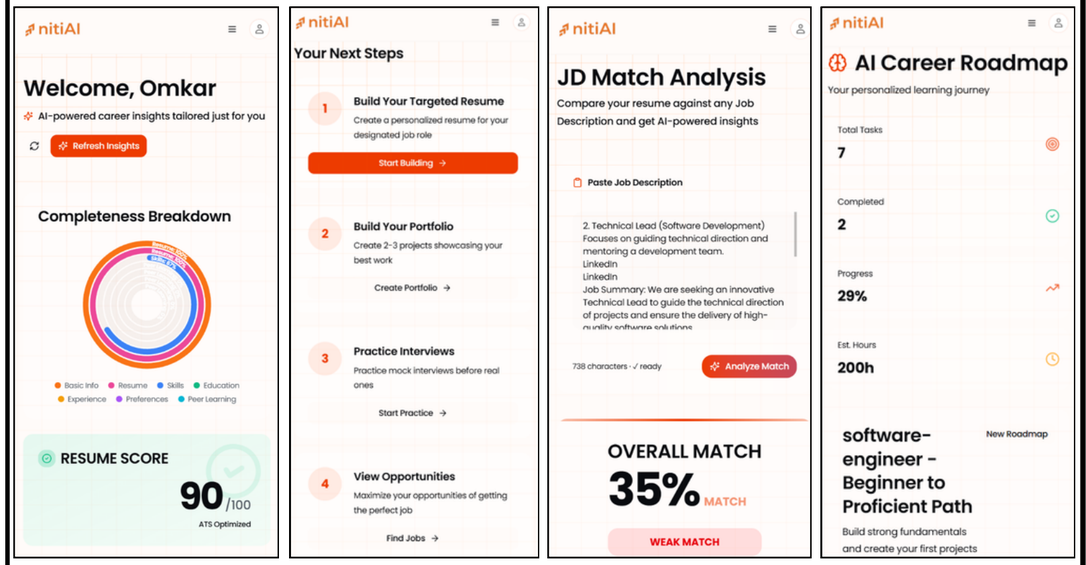
SYSTEM ARCHITECTURE



Tech Stack

redis · PostgreSQL · REST API · python · NGINX

PROTOTYPE



1.1 Create Your Profile → 1.2 Build Targeted Candidate Portfolio → 1.3 Score Resume Against Dedicated JD → 1.4 Follow Your Personalized Roadmap

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FEASIBILITY AND VIABILITY

Technical Feasibility

Microservices built using **Python/FastAPI** with REST APIs, semantic resume–JD matching powered by **NLP embeddings and contextual similarity scoring algorithms**.

Usability And Scalability

Use of **Nginx** [*Load Balancing*], **Docker** *[Containerization]*, **Redis** *[Caching]*, **and Cron Job Scheduling** helps in *high scalability*.

Data - Compliance Feasibility

Integration with structured skill frameworks (NSQF, SSC standards, public job taxonomies) ensures standardized skill classification **.**

Potential Risks

- Inconsistent resume formats
- Skill ambiguity and synonym variations
- Data privacy concerns

Potential Challenges

- Semantic mapping complexity
- Bias risk and contextual misinterpretation

Strategies To Overcome

- Standardized skill ontology mapping
- Context-aware transformer models with human validation
- Encryption and secure data handling mechanisms

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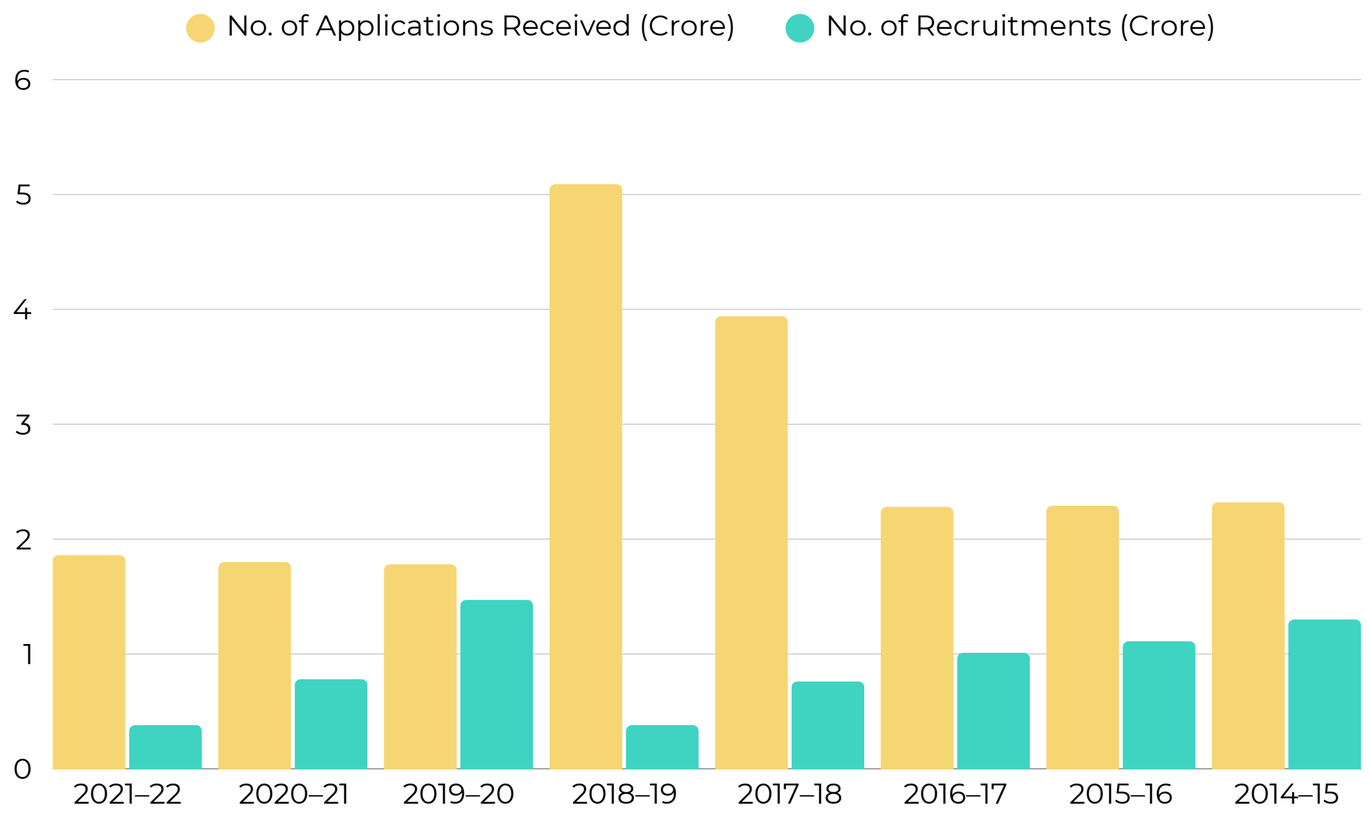
IMPACT AND BENEFITS



Number of government job applications in India surged from **2.29 crore (FY2016-17) to 1.87 crore (FY2021-22),** totaling over **22 crore** across 2014-22 against limited selections.

Candidates

Get accurate **skill-gap analysis, personalized improvement strategies,** and **structured career roadmaps**

Recruiters

Standardized resume scoring, automated filtering, semantic skill validation, and reduced manual screening effort.

Industry & EdTech Platforms

Access **structured skill intelligence, workforce benchmarking insights**, and **demand–supply skill gap analytics.**

Government

Real-time aggregated skill data enables **national skill-gap identification, workforce planning, and policy-driven upskilling initiatives**